Krishna Prasad Raveendran

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Education

University of Michigan, Ann Arbor, USA Expected May 2028
MS, Mechanical Engineering

National Institute of Technology Tiruchirappalli, India Jul 2019 – May 2023
B.Tech, Mechanical Engineering (First Class with Distinction) *CGPA: 8.94/10*

Conference Presentation

PyBaMM Battery Modelling Conference 2026, University of Michigan, Ann Arbor Oct 2026
Poster: DFN Parameterization and Validation of a High-Energy NCMA | Gr-SiOx 21700 Cell

- Open-source PyBaMM workflow from teardown to pack drive cycle: 9.12 mV OCV, 15 mV C-rate, 14.8 mV drive-cycle RMSE.

Experience

Graduate Research Assistant @ Battery Control Group, University of Michigan Aug 2026 – Present
State of charge estimation in silicon–graphite batteries: a unified framework accounting for degradation modes and hysteresis
Advisor: Prof. Anna Stefanopoulou

- Developing an SOC estimation framework for Si–Gr cells that captures silicon charge/discharge voltage hysteresis.
- Tracking degradation modes (LLI, LAM in graphite and silicon) so SOC estimates remain accurate as the cell ages.

Associate Cell Modeling Engineer @ EMO Energy, India Mar 2025 – Aug 2026

- Built DFN/P2D models for NCA/NMCA cells with Gr-SiOx anodes, validated to <15 mV RMSE at 0.2C–1C and 25/45 °C.
- Parameterized electrodes via 21700 teardown, coin cell rebuild, GITT and PITT into an in-house property database.
- Scaled validated cell models to pack level and verified response against UDDS and field-recorded drive cycles.
- Ran HPPC, DCIR, EIS and aging tests on 21700 and LFP cells, building a 14-month dataset for a 2,000-pack SOH model.
- Converted model outputs into BMS calibration tables and staged charging protocols, validated via HIL testing.

Cell Modeling – Product Analyst @ SwitchUp Technologies, India Jun 2024 – Jan 2025
Founding technical hire at a stealth startup accelerating EV adoption through battery financing and leasing

- Estimated SOH from partial slow-charge data on 2W/4W packs using 3.7–4.0 V CC-segment features, without a full RPT.
- Built a residual value model mapping RUL to second-life price bands, recovering 15% more value than fixed depreciation.
- Shipped the diagnostics and prognostics stack as an MVP, ingesting BMS telemetry from 100 vehicles in an Ather Energy pilot.

Cell Modeling and Design Engineer @ Log9 Materials, India Feb 2023 – Apr 2024

- Developed DFN/P2D and reduced-order models in PyBaMM and GT-Autolion for LFP cells in 66160 and prismatic formats.
- Replaced literature parameters with measured inputs from SEM-EDS, PSD, STA and AEM across teardown and test teams.
- Calibrated SEI, cracking, LAM and Li-plating degradation to measured SOH over a 9-point temperature/C-rate/DOD matrix.
- Optimized electrode composition, coating thickness and porosity for energy density vs. fast-charge trade-offs.

Summer Research Intern @ Fraunhofer IWM, Freiburg, Germany May 2022 – Aug 2022
Nano-droplet wetting effects on amorphous carbon using molecular dynamics

- Wrote LAMMPS scripts and ran nano-droplet wetting simulations on institute HPC as a fully repeatable workflow.
- Developed a novel contact angle algorithm from droplet density profiles, validated within 5% of DLIP experiments.

Technical Skills

Modeling & Simulation: GT-Autolion, PyBaMM, MATLAB, Simulink, LAMMPS, BPX (Battery Parameter Exchange)
Programming & Data: Python, NumPy, Pandas, Matplotlib, SciPy, Shell Scripting, Git, ML Applications
Experimental & Lab: Coin Cell Fabrication, Glovebox, HPPC, DCIR, EIS, GITT/PITT, SEM-EDS, Arbin/Neware Cyclers
Domain Expertise: DFN/P2D Electrochemical Modeling, MSMR OCP Modeling, State Estimation, Degradation Analysis

Achievements & Leadership

- **Fundraising Lead, PyBaMM (Nov 2024 – Present):** Developing fundraising strategy for the NumFOCUS-sponsored project.
- Awarded Summer Research Internship ’22 at Fraunhofer IWM, Freiburg, for atomistic friction simulation work.
- Recipient, Summer Research Fellowship Program ’22, Indian Academy of Sciences (national competitive selection).